

PEGAsus: 3D Personalization of Geometry and Appearance

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We present PEGAsus, a new framework capable of generating **P**ersonalized 3D shapes by learning shape concepts at both **G**eometry and **A**pppearance levels. First, we formulate 3D shape personalization as extracting reusable, category-agnostic geometric and appearance attributes from reference shapes, and composing these attributes with text to generate novel shapes. Second, we design a progressive optimization strategy to learn shape concepts at both the geometry and appearance levels, decoupling the shape concept learning process. Third, we extend our approach to region-wise concept learning, enabling flexible concept extraction, with context-aware and context-free losses. Extensive experimental results show that PEGAsus is able to effectively extract attributes from a wide range of reference shapes and then flexibly compose these concepts with text to synthesize new shapes. This enables fine-grained control over shape generation and supports the creation of diverse, personalized results, even in challenging cross-category scenarios. Both quantitative and qualitative experiments demonstrate that our approach outperforms existing state-of-the-art solutions.

1 Introduction

Creating distinctive 3D assets is a tedious and expensive process. Often, significant iterations are required to create a desired 3D design. So, after investing huge effort in making a design, artists typically seek to reuse a design’s geometric/appearance attributes to create some other designs, rather than creating new ones completely from scratch. Yet, such reusability is hard to achieve, especially for general 3D objects. To reuse geometric/appearance attributes requires not only the capability to identify and extract geometric/appearance attributes in the reference object but also the capability to incorporate the attributes to create new designs with controllability.

Existing methods struggle to achieve such a goal. Classical example-based 3D modeling methods [Bokeloh et al. 2010; Chaudhuri et al. 2011; Funkhouser et al. 2004] generally reuse a reference shape by assembling retrieved parts. This approach is essentially heuristic and operates at part level, offering limited semantic control in attribute extraction and incorporation. 3D stylization [Hu et al. 2017; Li et al. 2013; Xu et al. 2010] is another closely related approach, aiming to transfer the style of a reference shape to modify a given target shape. Yet, the approach assumes a given target shape. It *cannot* directly generate new shapes guided solely by the reference. This reliance limits its usability in 3D asset design, where artists

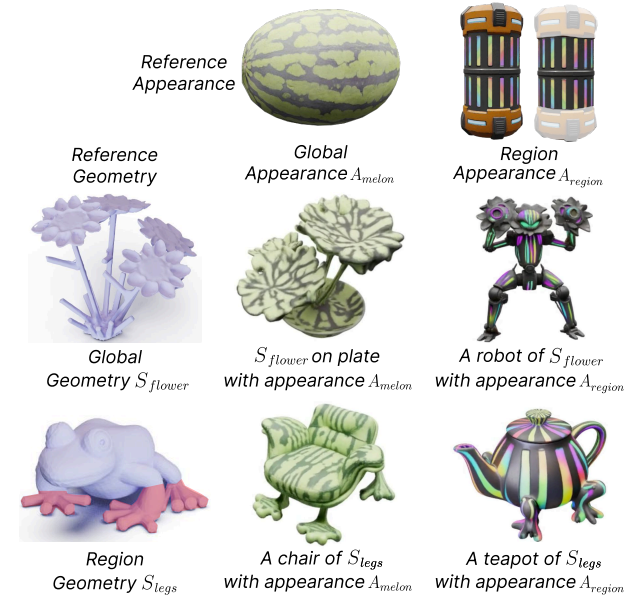


Fig. 1. PEGAsus is capable of extracting shape concepts for the global appearance of a watermelon, region-wise appearance of a stripe pattern, global geometry of a flower, and region-wise geometry of frog legs, then composing these learned concepts with text prompts in different ways to synthesize the four new shapes shown above, *e.g.*, a chair whose lower part follows the frog legs and appearance follows the watermelon.

often want to create new designs, even across shapes of different categories, without requiring efforts to pre-model the target.

In this work, we pursue 3D personalization. We aim to learn attributes from references to guide the shape generation, without prescribing a target instance. While this paradigm has been explored in image generation [Gal et al. 2023; Kumari et al. 2023; Ruiz et al. 2023], extending it to 3D presents unique challenges. Unlike 2D images, where attributes are entangled, 3D asset design requires explicitly decoupling the attribute controls over geometry and appearance, while supporting region-wise concept learning and cross-category reusability; see Figure 1 for a preview of our results.

In this paper, we introduce PEGAsus, short for 3D **P**ersonalization of **G**eometry and **A**pppearance (PEGA), reflecting its focus on learning reusable shape concepts applicable beyond a single instance or category. From a reference shape, PEGAsus can effectively learn reusable and category-agnostic shape concepts, where each concept

PEGAsus: 几何与外观的三维个性化

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我们提出了 PEGAsus，这是一种能够通过几何形状与外观层面学习形状概念来生成个性化三维形状的新框架。首先，我们将三维形状个性化定义为：从参考形状中提取可复用、类别无关的几何与外观属性，并将这些属性与文本组合，以生成新颖形状。其次，我们设计了一种渐进式优化策略，在几何和外观层面学习形状概念，从而解耦形状概念学习过程。第三，我们将该方法扩展至区域级概念学习，通过结合有上下文和无上下文损失，实现灵活的概念提取。大量实验结果表明，PEGAsus 能够从广泛的参考形状中有效提取属性，并将这些概念与文本灵活组合，以合成新形状。这使得我们能够对形状生成进行细粒度控制，并支持创建多样化的个性化结果，即使是在具有挑战性的跨类别场景中也是如此。定量和定性实验均证明，我们的方法优于现有的最先进解决方案。

1 引言

创建具有独特性的3D资产是一个繁琐且昂贵的过程。通常需要进行大量迭代，才能创建出理想的三维设计。因此，在投入大量精力完成一项设计后，艺术家通常会寻求再利用该设计的几何/外观属性来创建其他设计，而不是完全从头开始创建新的设计。然而，实现这种再利用并不容易，尤其是对于通用三维对象而言。要再利用几何/外观属性，不仅需要具备识别和提取参考对象中几何/外观属性的能力，还需要具备将这些属性融入新设计并实现可控创建的能力。

现有方法难以实现这一目标。经典的基于示例的三维建模方法 [Bokeloh 等人

。2010; Chaudhuri 等, 2011; Funkhouser 等, 2004] 通常通过组装检索到的部件来再利用参考形状。这种方法本质上是启发式的，并且在部件级上运行，在属性提取和整合方面提供的语义控制有限。三维风格化 [Hu 等, 2017; Li 等, 2013; Xu 等, 2010] 是另一种密切相关的方法，旨在将参考形状的风格迁移到给定的目标形状上以对其进行修改。然而，该方法假设目标形状已给定。它无法仅由参考图引导直接生成新形状。这种依赖限制了其在三维资产设计中的可用性，而在三维资产设计中，艺术家

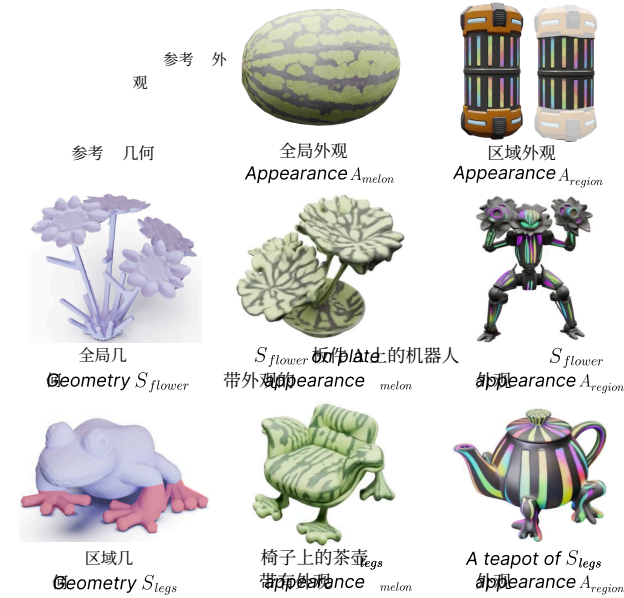


图1. PEGAsus能够提取西瓜的全局外观、条纹图案的区域外观、花朵的全局几何以及蛙腿的区域几何结构等形状概念，然后以不同方式将这些学习到的概念与文本提示组合起来，合成上方所示的四种新形状，例如，下半部分遵循蛙腿形状、外观遵循西瓜的椅子。

通常希望创建设计，甚至跨越不同类别的形状，而无需投入大量精力对目标进行预建模。

在这项工作中，我们致力于实现三维个性化。我们的目标是从参考图中学习属性，以引导形状生成，而不预先指定目标实例。尽管这一范式已在图像生成领域得到探索 [Gal 等, 2023; Kumari 等, 2023; Ruiz 等, 2023], 将其扩展到三维仍面临独特挑战。不同于属性彼此纠缠的二维图像，3D资产设计需要明确解耦对几何形状与外观的属性控制，同时支持区域级概念学习和跨类别复用性；我们的结果预览见图1。

在本文中，我们介绍了 PEGAsus，即 3D **P**ersonalization of **G**eometry and **A**pppearance (PEGA) 的缩写，体现了其专注于学习可复用的形状概念，这些概念适用于单个实例或类别之外的场景。给定一张参考形状，PEGAsus 能够有效地学习可复用且类别无关的形状概念，其中每个概念

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captures a specific geometric/appearance attribute. The learned concepts can then be composed with text to synthesize new shapes, while preserving the attributes extracted from the reference.

To achieve this, we build our framework on TRELLIS [Xiang et al. 2025], a large 3D foundation model, which naturally decouples geometry and appearance through a two-stage shape generation pipeline. Leveraging this model as a robust prior, we design a progressive optimization strategy to extract geometry- and appearance-level concepts from different stages of the pipeline, allowing their concepts to be learned independently. Further, PEGAsus supports both global concept learning from the entire shape and a region-wise mechanism that localizes the concept learning on a user-specified region. For region-wise concept learning, we introduce two complementary objectives: the context-aware loss to ensure visual coherence and the context-free loss to isolate the concept learning from the region’s surroundings. The learned concept is jointly represented by an optimized text embedding and a fine-tuned generator. At inference, the concept embedding can be composed with text and fed into the fine-tuned generator to synthesize new shapes.

PEGAsus supports four personalization modes: { global or region-wise concept learning } × { geometry or appearance }. As Figure 1 shows, PEGAsus is able to extract the global geometry concept of the flower, the region geometry concept of the frog’s legs, global appearance concept of the watermelon, and region appearance concept of the striped pattern, and then flexibly reuse and compose these concepts with texts to synthesize different new shapes.

In summary, we make the following contributions:

- We formulate the problem of *3D shape personalization* as extracting reusable, category-agnostic geometric and appearance attributes from reference shapes and composing the attributes with text to generate new shapes.
- We propose a concept-level 3D generation framework that independently learns and composes geometry-level and appearance-level shape concepts, while flexibly supporting both global and region-wise personalization.
- Both quantitative and qualitative results show that PEGAsus is able to produce a wide range of novel personalized 3D shapes with four distinct personalization modes, outperforming existing methods in both quality and flexibility.

2 Related Work

Prior related work spans several directions, exploring different ways of generating, editing, or adapting 3D content, from part-based synthesis and style propagation to large-scale generative modeling and concept learning. Below, we briefly review these areas and position our approach with respect to their assumptions and limitations.

Example-based 3D Modeling [Funkhouser et al. 2004] aims to synthesize new shapes by copying, deforming, and assembling parts from a database of exemplar shapes. Subsequent works [Bokeloh et al. 2010; Chaudhuri et al. 2011; Fisher et al. 2012; Kalogerakis et al. 2012; Merrell 2007; Merrell and Manocha 2008; Xu et al. 2012; Zhou et al. 2007] extend this paradigm with probabilistic modeling or part-based composition. Their reliance on manual decomposition or handcrafted priors, *e.g.*, symmetry, however, restricts the synthesis

to a recombination of database components, limiting the generalization to novel or cross-category scenarios. Recently, learning-based methods [Li et al. 2023; Maruani et al. 2025; Wu et al. 2024b; Wu and Zheng 2022] adopt generative models to implicitly extract regions or patches for producing plausible shape variants from a single reference shape. These approaches, however, focus primarily on exemplar-specific diversity, *e.g.*, modify the aspect ratio of the exemplar shape, offering limited semantic control over the generated shapes. In contrast, our method learns reusable, category-agnostic attributes from the reference shape, enabling more flexible semantic control. The extracted attributes can be composed with texts to synthesize new shapes, even for cross-category scenarios.

3D stylization. Given a reference style input and a target content shape, 3D stylization aims to extract the stylistic patterns of the reference and re-apply them to the target to synthesize new shapes. This approach operates across various representations, including explicit meshes [Hertz et al. 2020; Höllein et al. 2022; Hu et al. 2017; Kato et al. 2018; Li et al. 2013; Michel et al. 2022; Xu et al. 2010; Yin et al. 2021], neural implicit fields, and Gaussian splatting [Fan et al. 2022; Huang et al. 2021, 2022; Kumar et al. 2023; Liu et al. 2023b; Nguyen-Phuoc et al. 2022; Zhang et al. 2025, 2023a]. To enhance the generalizability, recent approaches [Chen et al. 2025; Dong et al. 2024; Xie et al. 2024] leverage strong 2D priors from large-scale image diffusion models [Rombach et al. 2022] for 3D stylization. Meanwhile, Qu et al. [2025] employs 3D priors [Xiang et al. 2025] for consistent stylization across views. Crucially, all these approaches follow a style-transfer formulation, requiring a user-provided target shape to define the geometry structure. In contrast, our method addresses a distinct task of reference-guided 3D shape generation. From the reference, rather than extracting explicit geometry, we extract category-agnostic shape concepts that can be composed flexibly with texts to synthesize new shapes, considering the structural semantics in the text, while preserving the learned geometric/appearance attributes from the reference shape.

3D Generative Modeling. Inspired by the success of diffusion models in 2D image and video generation [Ho et al. 2020, 2022], recent works [Cheng et al. 2023; Hu et al. 2024a; Liu et al. 2023a; Ren et al. 2024; Zeng et al. 2022; Zhang et al. 2023b] have significantly advanced 3D generative modeling by learning compact latent representations and performing generation in the compressed space. With the emergence of large-scale 3D datasets [Deitke et al. 2023a,b], 3D foundation models [He et al. 2025; Hong et al. 2025, 2024; Hui et al. 2022; Li et al. 2025; Wu et al. 2024a, 2025b; Xiang et al. 2025; Zhang et al. 2024; Zhao et al. 2025] have been developed to capture strong geometric priors supporting not only high-quality asset generation across diverse categories but also a wide range of downstream tasks, *e.g.*, shape analysis [Du et al. 2025] and editing [Hu et al. 2024b]. Despite their impressive generalization, these models primarily generate samples from a learned distribution and lack mechanisms for user-specified personalization or example-based control. In this work, we leverage the robust priors of the 3D foundation model TRELLIS [Xiang et al. 2025] with a decoupled personalization framework to enable flexible user-specified adaptation

都体现了一种特定的几何/外观属性。随后，可以将学习到的概念与文本结合，以合成新的形状，同时保留从参考图中提取的属性。

为此，我们基于 TRELLIS [Xiang et al.2025].这一大型 3D 基础模型构建了我们的框架。该模型通过两阶段形状生成管线，自然地将几何与外观解耦。我们将这一模型作为稳健的先验，并设计了一种渐进式优化策略，从流程的不同阶段提取几何层级概念和外观层级概念，使这些概念能够独立学习。此外，PEGAsus 同时支持从整个形状中进行全局概念学习，以及通过区域级机制在用户指定区域内定位概念学习。对于区域级概念学习，我们提出了两个互补目标：上下文感知损失用于确保视觉连贯性，而无上下文损失用于将概念学习与区域周围环境隔离。学习到的概念由经过优化的文本嵌入和微调生成器共同表示。在推理阶段，可以将概念嵌入与文本组合，并输入微调生成器，以合成新的形状。

PEGAsus 支持四种个性化模式：{ 全局或区域概念学习 } × { 几何或外观 }。如图 1 所示，PEGAsus 能够提取花朵的全局几何概念、青蛙腿的区域几何概念、西瓜的全局外观概念以及条纹图案的区域外观概念，然后将这些概念与文本灵活地再利用和组合，以合成不同的新形状。

总而言之，我们做出了以下贡献：

- 我们将< astyle id='2'>三维形状个性化 问题表述为：从参考形状中提取可再利用的、类别无关的几何和外观属性，并将这些属性与文本组合，以生成新的形状。
- 我们提出了一种概念级三维生成框架，能够独立学习并组合几何级和外观级形状概念，同时灵活支持全局和区域个性化。
- 定量和定性结果均表明，PEGAsus 能够生成广泛的新颖个性化

具有四种不同个性化模式的三维形状，在质量和灵活性方面均优于现有方法。

2 相关工作

以往相关工作涵盖多个方向，探索了生成、编辑或调整三维内容的不同方式，从基于部件的合成和风格传播，到大规模生成建模与概念学习。下面我们将简要回顾这些领域，并结合它们的假设与局限性，对我们的方法进行定位。

基于示例的三维建模 [Funkhouser 等，2004] 旨在通过从示例形状数据库中复制、变形和组装部件来合成新的形状。后续工作 [Bokeloh 等，2010; Chaudhuri 等，2011; Fisher 等，2012; Kalogerakis 等，2012; Merrell, 2007; Merrell 和 Manocha, 2008; Xu 等，2012; Zhou 等，2007] 通过概率建模或基于部件的组合扩展了这一范式。然而，这些方法依赖手工分解或手工设计的先验，例如对称性，限制了合成结果

仅限于数据库组件的重组，限制了其向新颖或跨类别场景的泛化能力。近年来，基于学习的方法 [Li 等，2023; Maruani 等，2025; Wu 等，2024b; Wu 和 Zheng 2022] 采用生成模型，隐式提取区域或图块，从单个参考形状生成合理的形状变体。然而，这些方法主要关注样例特定多样性，例如修改样例形状的纵横比，对生成形状提供的语义控制有限。相比之下，我们的方法从参考形状中学习可复用的、类别无关的属性，从而实现更灵活的语义控制。提取出的属性可以与文本组合，以合成新的形状，即使是在跨类别场景下也能实现。

三维风格化。给定参考风格输入和目标内容形状，三维风格化旨在提取参考图的风格模式，并将其重新应用于目标形状，以合成新的形状。这种方法适用于多种表示形式，包括显式网格 [Hertz 等，2020; Höllein 等，2022; Hu 等，2017; Kato 等，2018; Li 等，2013; Michel 等，2022; Xu 等，2010; Yin 等，2021], 神经隐式场，以及高斯泼溅 [Fan 等，2022; Huang 等，2021, 2022; Kumar 等，2023; Liu 等，2023b; Nguyen-Phuoc 等，2022; Zhang 等，2025, 2023a]。为增强泛化能力，近期方法 [Chen 等，2025; Dong 等，2024; Xie 等，2024] 利用大规模图像扩散模型 [Rombach 等，2022] 提供的强二维先验，实现三维风格化。同时，Qu 等，[2025] 采用三维先验 [Xiang 等，2025]，以实现跨视角的一致风格化。重要的是，所有这些方法都遵循风格迁移范式，需要用户提供目标形状来定义几何结构。相比之下，我们的方法解决的是一个不同的任务，即参考引导的三维形状生成。我们并非从参考图中提取显式几何，而是提取类别无关的形状概念，并将其与文本灵活组合，以合成新的形状，

考虑文本中的结构语义，同时保留从参考形状中学习到的几何/外观属性

三维生成建模。受扩散模型在二维图像和视频生成领域取得成功的启发 [Ho 等人 2020, 2022], 近期研究 [Cheng 等，2023; Hu 等人 2024a; Liu 等人 2023a; Ren 等人 2024; Zeng 等人 2022; Zhang 等人 2023b] 通过学习紧凑的潜变量表示并在压缩空间中执行生成，显著推动了三维生成建模的发展。随着大规模

3 [2023].D 数据集 Deitke 等人 a、b 以及 D 基础模型 He 等人; Hong 等人; Hui 等人 2022; Li 等人 2025; Wu 等人 2024a、2025b; Xiang 等，2025; Zhang 等人 2024; Zhao 等人 2025] 的出现，人们开发出了能够捕捉强大几何先验的模型，不仅支持跨不同类别的高质量资产生成，还支持广泛的下游任务，例如形状分析 [Du 等人 2025] 和编辑 [Hu 等人 2024b]。尽管这些模型具有出色的泛化能力，但它们主要从学习到的分布中生成样本，缺乏用户指定的个性化或基于示例的控制机制。在本文中，我们利用三维基础模型 TRELLIS [Xiang 等人 2025] 的稳健先验，并结合解耦的个性化框架，实现灵活的用户指定适应

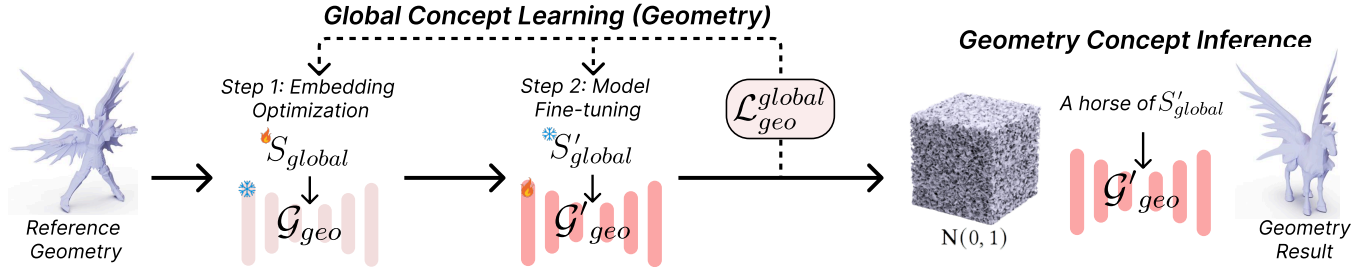


Fig. 2. Global concept learning in PEGAsus. Given a reference shape, we first optimize the text embedding S_{global} to S'_{global} then fine-tune the geometry generator $\mathcal{G}_{geo}$ to $\mathcal{G}'_{geo}$; both steps are guided by the global (geometry) loss $\mathcal{L}_{geo}^{global}$. At inference, we compose the learned concept S'_{global} with a text prompt and use the fine-tuned model to generate new shapes with the inherited reference attributes. This figure shows global concept learning for geometry, yet the same pipeline can be used for appearance, by replacing $\mathcal{G}_{geo}$ with the appearance generator and adjusting the reference data from geometry to appearance.

of geometry and appearance, while maintaining the high-quality generative capabilities of the base model.

Image personalization aims to adapt large text-to-image generative models [Rombach et al. 2022] to capture the visual identity of a specific subject or concept from a set of reference images, enabling faithful synthesis under new text prompts. Textual Inversion [Gal et al. 2023] introduces personalization by learning a concept-specific text embedding, whereas DreamBooth [Ruiz et al. 2023] improves the fidelity via model fine-tuning. Subsequent methods [Gu et al. 2023; Kumari et al. 2023; Ruiz et al. 2024; Safaei et al. 2024; Shi et al. 2024; Tewel et al. 2023; Wei et al. 2023; Ye et al. 2023] increase efficiency and controllability by updating only lightweight parameters or leveraging image encoders, and others [Avrahami et al. 2023; Safaei et al. 2024] further extend personalization to spatially localized concepts for fine-grained, compositional control. Despite these advances, direct 3D personalization remains underexplored. Existing works [Huang et al. 2025; Liu et al. 2024; Raj et al. 2023; Wang et al. 2024] propose to lift personalized image diffusion priors into 3D, so the geometry and appearance are entangled, with limited geometric control and weak cross-category generalization. In contrast, we perform personalization directly in 3D, explicitly disentangling geometry- and appearance-level concepts, enabling cross-category concept-level shape generation. Further, we formulate a region-wise mechanism to localize the concept learning.

3 Overview

We develop the shape concept learning pipelines in PEGAsus based on the geometry generator $\mathcal{G}_{geo}$ and appearance generator $\mathcal{G}_{app}$ of TRELLIS [Xiang et al. 2025]; see Section 4.1. Importantly, we model the shape concept *jointly* by a *learnable text embedding* and a *generator model* fine-tuned from $\mathcal{G}_{geo}$ or $\mathcal{G}_{app}$. Below, we first overview the global concept learning pipeline in PEGAsus then introduce how we adapt it for region-wise concept learning.

Figure 2 shows the pipeline for global concept learning. Here, we focus on geometry concept, as appearance concept follows a similar procedure. Given a reference shape, we first adopt a progressive optimization strategy (Section 4.2) to extract a global concept. Then, we optimize the text embedding S_{global} to S'_{global} with a frozen generator ($\mathcal{G}_{geo}$, for the case of geometry concept) to extract coarse global attribute, without disrupting the generator’s prior. Next, we fine-tune generator $\mathcal{G}_{geo}$ to $\mathcal{G}'_{geo}$ to adapt it to fine-grained attribute

details, while fixing the optimized text embedding S'_{global} . Crucially, both steps employ the same objective $\mathcal{L}_{geo}^{global}$, differing only in the parameters being optimized. At inference, the learned geometry concept embedding S'_{global} can be composed with text and fed into the fine-tuned generator $\mathcal{G}'_{geo}$ to synthesize new shapes (Section 4.4).

Figure 3 shows the pipeline for region-wise concept learning. For brevity, we illustrate it with appearance concept, as geometry concept follows analogously. Given a reference shape with a user-specified region, we first extract region-wise concept using the progressive optimization strategy. Then, we adapt the global pipeline by introducing two complementary losses, *i.e.*, context-aware loss $\mathcal{L}^{ctx}$ and context-free loss $\mathcal{L}^{free}$, to facilitate independent yet coherent region-wise concept learning (Section 4.3). At inference, the learned appearance concept can then be employed to generate shapes with the desired appearance attributes (Section 4.4).

4 Method

4.1 Preliminary: TRELLIS

First, we briefly review TRELLIS [Xiang et al. 2025] for the necessary background. TRELLIS uses a unified *Structured Latent* (SLAT) representation $\{(V, Z) \mid V = \{p_i\}_{i=1}^L, Z = \{z_i\}_{i=1}^L\}$, where V defines the geometry structure via spatial coordinates p_i and Z defines the associated appearance features z_i . This representation decouples the 3D generation into two stages: (i) a sparse structure stage to synthesize V and (ii) a structured latent stage to generate Z .

Stage 1: Sparse Structure Stage. This stage encodes geometry V into a latent voxel grid X using a pre-trained VAE, then trains the geometry generator $\mathcal{G}_{geo}$ to denoise X , conditioned on text y :

$$\mathcal{L}_{geo} = \mathbb{E}_{t, X_t, y} [\|v_t - \mathcal{G}_{geo}(X_t, t, y)\|^2], \quad (1)$$

where X_t is the noisy latent at time t and v_t is the target velocity. Then, the VAE decoder maps the denoised X to generate V .

Stage 2: Structured Latent Stage. Given the geometry V , the second stage trains the appearance generator $\mathcal{G}_{app}$ to denoise $Z = \{z_i\}_{i=1}^L$ from Gaussian noise, conditioned on text y and the fixed V :

$$\mathcal{L}_{app} = \mathbb{E}_{t, Z_t, y, V} [\|u_t - \mathcal{G}_{app}(Z_t, t, y, V)\|^2], \quad (2)$$

where Z_t is the noisy latent at time t and u_t is the target velocity. The resulting structured latent (V, Z) can then be decoded into various 3D representations, *e.g.*, 3D Gaussian splats and meshes.

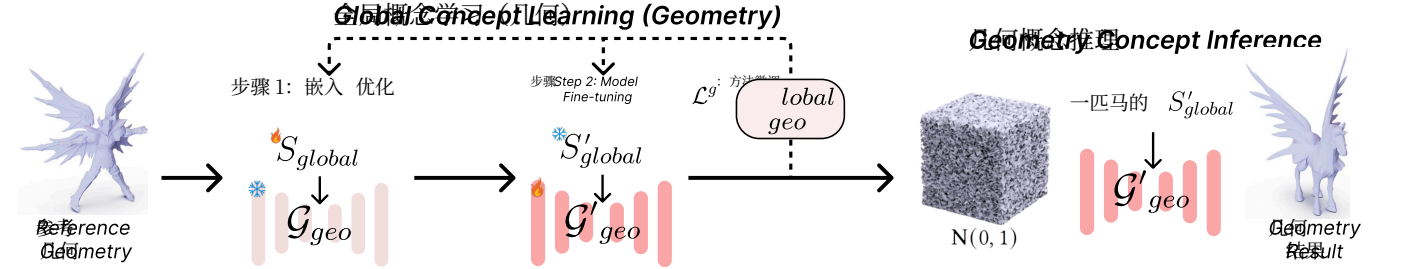


图 2. PEGAsus 中的全局概念学习。给定一个参考形状，我们首先优化文本嵌入 S_{global} 以 S'_{global} ，然后将几何生成器 $\mathcal{G}_{geo}$ 微调为 $\mathcal{G}'_{geo}$ ；这两个步骤都由全局（几何）损失 $\mathcal{L}_{geo}^{global}$ 引导。在推理阶段，我们将学习到的概念 S'_{global} 与文本提示组合起来，并使用微调后的模型生成继承参考属性的新形状。该图展示了几何的全局概念学习，但通过将 $\mathcal{G}_{geo}$ 替换为外观生成器，并将参考数据从几何调整为外观，也可以使用相同的流程进行外观学习。

同时保持基础模型的高质量生成能力，实现对几何形状与外观的适应。

图像个性化旨在调整大型文本到图像生成模型 [Rombach 等 2022]，使其能够从一组参考图像中捕捉特定主体或概念的视觉身份，从而在新的文本提示下实现忠实合成。文本反演 [Gal 等 2023] 通过学习特定概念文本嵌入引入了个性化，而 DreamBooth [Ruiz 等 2023] 则通过模型微调提高了保真度。后续方法 [Gu 等, 2023; Kumari 等, 2023; Ruiz 等, 2024; Safaei 等, 2024; Shi 等, 2024; Tewel 等, 2023; Wei 等, 2023; Ye 等 2023] 通过仅更新轻量级参数或利用图像编码器来提高效率和可控性，其他方法 [Avrahami 等, 2023; Safaei 等 2024] 则进一步将个性化扩展到空间局部化概念，以实现细粒度的组合控制。尽管取得了这些进展，直接的三维个性化仍未得到充分探索。现有工作 [Huang 等, 2025; Liu 等, 2024; Raj 等, 2023; Wang 等 2024] 提出将个性化图像扩散先验提升到三维空间，因此几何形状与外观彼此纠缠，几何控制能力有限，且跨类别泛化能力较弱。相比之下，我们直接在三维空间中执行个性化，明确解耦几何层级概念与外观层级概念，从而实现跨类别概念级形状生成。此外，我们设计了区域级机制，以定位概念学习。

3 概述

我们基于 TRELLIS [Xiang 等人 2025] 的几何生成器 $\mathcal{G}_{geo}$ 和外观生成器 $\mathcal{G}_{app}$ ，在 PEGAsus 中构建形状概念学习流程；详见第 4.1 节。重要的是，我们通过可学习的文本嵌入和生成器模型，对从 $\mathcal{G}_{geo}$ 或 $\mathcal{G}_{app}$ 微调得到的形状概念进行联合建模。下文首先概述 PEGAsus 中的全局概念学习流程，然后介绍如何将其调整用于区域级概念学习。

图 2 展示了全局概念学习的流程。这里我们重点介绍几何概念，因为外观概念遵循类似的过程。给定一个参考形状，我们首先采用渐进式优化策略（第 4.2 节）来提取全局概念。然后，在冻结生成器 ($\mathcal{G}_{geo}$ ，的情况下，我们优化文本嵌入 S_{global} 至 S'_{global} （针对几何概念的情况），以提取粗粒度全局属性，同时不破坏生成器的先验。接下来，我们将生成器 $\mathcal{G}_{geo}$ 微调为 $\mathcal{G}'_{geo}$ ，使其适应细粒度属性

细节，同时固定优化后的文本嵌入 S'_{global} 。关键的是，这两个步骤采用相同的目标函数 $\mathcal{L}_{geo}^{global}$ ，区别仅在于所优化的参数不同。在推理时，学习到的几何概念嵌入 S'_{global} 可以与文本组合，并输入微调后的生成器 $\mathcal{G}'_{geo}$ ，以合成新的形状（第 4.4 节）。

图 3 展示了区域级概念学习的流程。为简洁起见，我们以外观概念为例进行说明，因为几何概念遵循类似的过程。给定一个带有用户指定区域的参考形状，我们首先使用渐进式优化策略提取区域级概念。然后，我们通过引入两个互补损失来调整全局流程，即上下文感知损失 $\mathcal{L}^{ctx}$ 和无上下文损失 $\mathcal{L}^{free}$ ，从而促进独立但协调一致的区域级概念学习（第 4.3 节）。在推理阶段，学习到的外观概念随后可用于生成具有所需外观属性的形状（第 4.4 节）。

4 方法

4.1 初步介绍: TRELLIS

首先，我们简要回顾 TRELLIS [Xiang 等人 2025]，以介绍必要的背景知识。TRELLIS 使用统一的 *结构化潜变量* (SLAT) 表示 $\{(V, Z) \mid V = \{p_i\}_{i=1}^L, Z = \{z_i\}_{i=1}^L\}$ ，其中 V 通过空间坐标定义几何结构 p_i ，而 Z 定义相关的外观特征 z_i 。该表示将三维生成解耦为两个阶段：(i) 稀疏结构阶段，用于合成 V ；以及 (ii) 结构化潜变量阶段，用于生成 Z 。

阶段 1：稀疏结构阶段。 该阶段使用预训练变分自编码器将几何 V 编码为潜在体素网格 X ，随后训练几何生成器 $\mathcal{G}_{geo}$ 对 X 进行去噪，并以文本 y 为条件：

$$\mathcal{L}_{geo} = \mathbb{E}_{t, X_t, y} [\|v_t - \mathcal{G}_{geo}(X_t, t, y)\|^2], \quad (1)$$

其中， X_t 是时间 t 下的带噪潜变量， v_t 是目标速度。随后，变分自编码器解码器将去噪后的 X 映射为 V 。

第二阶段：结构化潜变量阶段。 给定几何 V ，第二阶段训练外观生成器 $\mathcal{G}_{app}$ 对 $Z = \{z_i\}_{i=1}^L$ 进行去噪，使其从高斯噪声中恢复，并以文本 y 和固定的 V 为条件：

$$\mathcal{L}_{app} = \mathbb{E}_{t, Z_t, y, V} [\|u_t - \mathcal{G}_{app}(Z_t, t, y, V)\|^2], \quad (2)$$

其中， Z_t 是时间 t 的带噪潜变量， u_t 是目标速度。由此得到的结构化潜变量 (V, Z) 随后可以解码为各种三维表示，例如三维高斯点和网格。

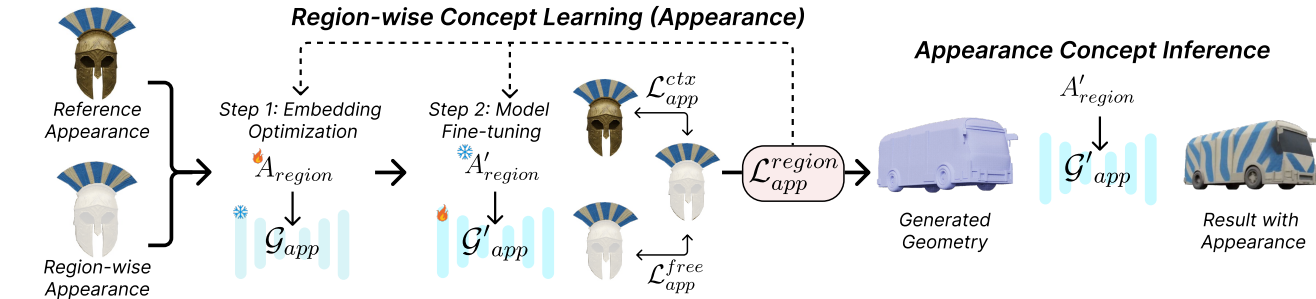


Fig. 3. Region-wise concept learning in PEGAsus. Given a reference appearance with a user-specified region, we perform a progressive optimization strategy with a region-wise (appearance) objective $\mathcal{L}_{app}^{region}$. This objective combines $\mathcal{L}_{app}^{ctx}$ and $\mathcal{L}_{app}^{free}$ to encourage the learned concept to be visually coherent and conceptually independent. At inference, the concept synthesizes result that inherits the attributes of the specified region. While this figure shows region-wise concept learning for appearance, the pipeline applies to geometry by replacing G_{app} with G_{geo} and switching reference data to geometry.

Building on this SLAT representation, we treat geometry and appearance as distinct personalization targets, facilitating decoupled concept learning at both the geometry and appearance levels.

4.2 Global Concept Learning

Fundamentally, 3D personalization involves a trade-off between (i) concept fidelity, *i.e.*, how accurately the concept captures the reference attributes, and (ii) transferability, *i.e.*, how well the concept can be composed with text to generate new shapes. Relying solely on optimizing text embeddings is often insufficient, as the compact text embedding space lacks the capacity to encode detailed geometric or appearance attributes; see an ablation in Figure 7(c). Meanwhile, directly fine-tuning the generator on a single reference shape usually disrupts its 3D priors, leading to overfitting where the fine-tuned model fails to generalize to novel shapes; see Figure 7(d).

Progressive Optimization Strategy. To address the above issues, we formulate the progressive optimization strategy to optimize the joint shape concept, *i.e.*, a text embedding and a generator model. Below, we denote $\mathcal{G}$ as the generator, $\mathcal{D}$ the reference shape representation, and C a learnable text embedding initialized from a neutral text description, *e.g.*, “object”, encoded by CLIP [Radford et al. 2021].

- **Step 1: Embedding Optimization.** We first capture coarse global attributes by optimizing only the text embedding. Since the text embedding C is initialized from a neutral word, we utilize a relatively high learning rate to optimize it to C' while freezing the generator $\mathcal{G}$. This encourages the generic text embedding to align with the coarse global structure $\mathcal{D}$ without altering the generative prior.
- **Step 2: Model Fine-tuning.** To recover fine-grained details that the embedding alone cannot represent, we fix the optimized text embedding C' and fine-tune generator $\mathcal{G}$ to $\mathcal{G}'$ using a low learning rate. This conservative update alleviates overfitting to the single reference instance while allowing the model to capture high-frequency details of $\mathcal{D}$ that were missed in the embedding optimization step.

Crucially, while the two steps have different optimization parameters, the objective function remains identical. Below, we specify the reference data $\mathcal{D}$, the learnable text embedding C , the generator $\mathcal{G}$, and the objective function for each case of concept learning.

Global Geometry Concept Learning. For geometry concept learning, we construct the reference shape $\mathcal{D}$ by first voxelizing the reference shape, and encoding the voxel grid into the latent feature grid X . Then, we initialize the learnable text embedding C as S_{global} and optimize it to S'_{global} . We set the generator $\mathcal{G}$ as $\mathcal{G}_{geo}$, adopt the same flow matching objective as Equation 1, but replace the standard text condition y with our learnable text embedding S_{global} :

$$\mathcal{L}_{geo}^{global} = \mathbb{E}_{t, X_t} [\|v_t - \mathcal{G}_{geo}(X_t, t, S_{global})\|^2]. \quad (3)$$

Global Appearance Concept Learning. We define the appearance reference $\mathcal{D}$ as the SLAT representation (V, Z) extracted from the reference shape, following TRELLIS [Xiang et al. 2025], where Z serves as the target appearance features and V as the fixed geometry condition. We set the generator $\mathcal{G}$ as $\mathcal{G}_{app}$, initialize the learnable text embedding C as A_{global} , and then optimize it to A'_{global} . The global appearance loss follows Equation 2, but replacing the text prompt y with the appearance text embedding A_{global} :

$$\mathcal{L}_{app}^{global} = \mathbb{E}_{t, Z_t, V} [\|u_t - \mathcal{G}_{app}(Z_t, t, A_{global}, V)\|^2]. \quad (4)$$

The final learned concept is jointly represented by the optimized embedding and the fine-tuned generator: $(S'_{global}, \mathcal{G}'_{geo})$ for global geometry concept and $(A'_{global}, \mathcal{G}'_{app})$ for global appearance concept.

4.3 Region-wise Concept Learning

While global concept learning effectively captures the overall attributes of a reference shape, it lacks the precision to extract concepts from a specific local region. To enhance user controllability and enable more flexible personalization, we introduce region-wise concept learning, which enables the extraction of geometry and appearance concepts from a user-specified region. Given a reference shape $\mathcal{D}$ and a mask M denoting a user-specified region, we learn a region-wise shape concept $(C_{region}, \mathcal{G})$, to capture the attributes within M . To achieve this, we employ the same progressive optimization strategy proposed in the Section 4.2.

However, a challenge in region-wise concept learning is balancing context awareness with concept independence. When optimizing using the full reference shape as input while restricting supervision solely to the masked region, the generator tends to exploit the surrounding context to infer the masked region. This leads to entanglement, where the learned concept relies on the global shape

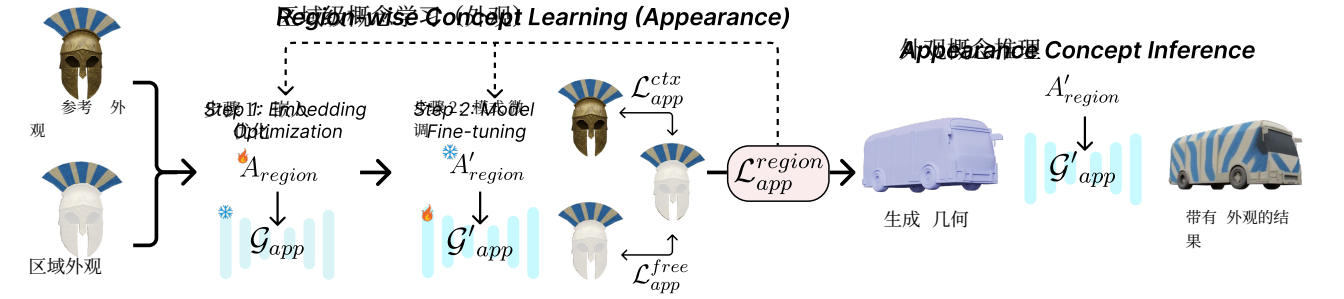


图 3. PEGAsus 中的区域级概念学习。给定一张包含用户指定区域的参考外观，我们采用带有区域级（外观）目标的渐进式优化策略 $\mathcal{L}_{app}^{region}$ 。该目标结合了 $\mathcal{L}_{app}^{ctx}$ 和 $\mathcal{L}_{app}^{free}$ ，以促使学习到的概念在视觉上保持连贯，并在概念上相互独立。在推理阶段，该概念合成继承指定区域属性的结果。虽然本图展示的是用于外观的区域级概念学习，但将 G_{app} 替换为 G_{geo} 并将参考数据切换为几何后，该流程同样适用于几何。

基于这一 SLAT 表示，我们将几何与外观视为相互独立的个性化目标，从而促进在几何和外观层面进行解耦概念学习。

4.2 全局概念学习

从根本上说，三维个性化涉及以下两者之间的权衡：(i) 概念保真度，即概念对参考属性的捕捉准确程度；以及 (ii) 可迁移性，即概念与文本组合以生成新颖形状的能力。仅依赖优化文本嵌入通常是不够的，因为紧凑的文本嵌入空间缺乏编码详细几何或外观属性的能力；消融实验见图7(c)。与此同时，直接针对单个参考形状微调生成器通常会破坏其三维先验，导致过拟合，使微调后的模型无法泛化到新颖形状；见图7(d)。

渐进式优化策略。为解决上述问题，我们提出渐进式优化策略，以优化联合形状概念，即文本嵌入和生成器模型。下文中，我们将 $\mathcal{G}$ 记作生成器，将 $\mathcal{D}$ 记作参考形状表示，并将 C 记作一个可学习的文本嵌入；该嵌入从中性文本描述（例如“object”）初始化，并由 CLIP [Radford et al. 2021] 进行编码。

- **步骤 1：嵌入优化。**我们首先仅通过优化文本嵌入来捕获粗粒度全局属性。由于文本嵌入 C 从中性词初始化，我们采用相对较高的学习率对其进行优化，使其变为 C' ，同时冻结生成器 $\mathcal{G}$ 。这促使通用文本嵌入与粗粒度全局结构 $\mathcal{D}$ 对齐，而不会改变生成先验。

- **步骤 2：模型微调。**为了恢复仅凭嵌入无法表示的细粒度细节，我们固定优化后的文本嵌入 C' ，并使用较低的学习率将生成器 $\mathcal{G}$ 微调为 $\mathcal{G}'$ 。这种保守的更新方式减轻了对单个参考实例的过拟合，同时使模型能够捕获嵌入优化步骤中遗漏的 $\mathcal{D}$ 的高频细节。

关键在于，尽管这两个步骤具有不同的优化参数，但目标函数保持不变。下面，我们分别说明概念学习各个情况下的参考数据 $\mathcal{D}$ 、可学习的文本嵌入 C 、生成器 $\mathcal{G}$ 以及目标函数。

全局几何概念学习。对于几何概念学习，我们首先将参考形状进行体素化，并将体素网格编码为潜在特征网格 X ，从而构造参考形状 $\mathcal{D}$ 。然后，我们将可学习的文本嵌入 C 初始化为 S_{global} ，并对其进行优化，使其达到 S'_{global} 。我们将生成器 $\mathcal{G}$ 设为 $\mathcal{G}_{geo}$ ，采用与公式 1 相同的流匹配目标，但将标准文本条件 y 替换为我们的可学习文本嵌入 S_{global} ：

$$\mathcal{L}_{geo}^{global} = \mathbb{E}_{t, X_t} [\|v_t - \mathcal{G}_{geo}(X_t, t, S_{global})\|^2]. \quad (3)$$

全局外观概念学习。我们将外观参考图 $\mathcal{D}$ 定义为从参考形状中提取的 SLAT 表示 (V, Z) ，遵循 TRELLIS [Xiang 等人 2025]，其中 Z 作为目标外观特征， V 作为固定的几何条件。我们将生成器 $\mathcal{G}$ 设为 $\mathcal{G}_{app}$ ，将可学习的文本嵌入 C 初始化为 A_{global} ，然后将其优化为 A'_{global} 。全局外观损失遵循公式 2，但将文本提示 y 替换为外观文本嵌入 A_{global} ：

$$\mathcal{L}_{app}^{global} = \mathbb{E}_{t, Z_t, V} [\|u_t - \mathcal{G}_{app}(Z_t, t, A_{global}, V)\|^2]. \quad (4)$$

最终学习到的概念由优化嵌入和微调生成器共同表示： $(S'_{global}, \mathcal{G}'_{geo})$ 用于全局几何概念， $(A'_{global}, \mathcal{G}'_{app})$ 用于全局外观概念。

4.3 区域级概念学习

尽管全局概念学习能够有效捕捉参考形状的整体属性，但缺乏从特定局部区域中提取概念的精确性。为增强用户可控性并实现更灵活的个性化，我们引入区域级概念学习，从用户指定区域中提取几何形状与外观概念。给定参考形状 $\mathcal{D}$ 和表示用户指定区域的掩码 M ，我们学习区域级形状概念 $(C_{region}, \mathcal{G})$ ，以捕捉 M 内的属性。为此，我们采用第 4.2 节提出的相同渐进式优化策略。

然而，区域级概念学习中的一个挑战是平衡上下文感知与概念独立性。当以完整参考形状作为输入进行优化，同时将监督严格限制在掩码区域内时，生成器往往会利用周围上下文来推断掩码区域。这会导致纠缠，使学习到的概念依赖于全局形状

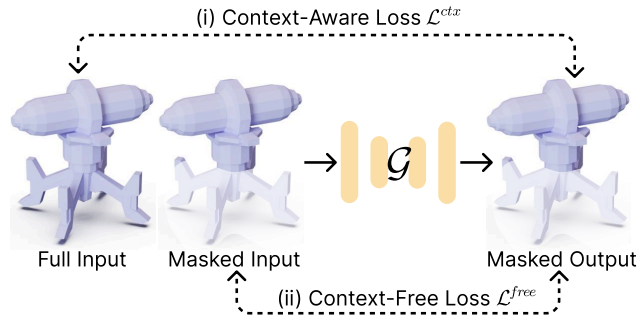


Fig. 4. Region-wise concept learning objectives. (i) Context-Aware Loss takes full reference and computes loss within the masked region. (ii) Context-Free Loss uses masked input to calculate loss within the masked region.

rather than being an independent attribute. Conversely, if we strictly isolate the region by masking out the rest of the shape, the concept may lose its semantic coherence with the global geometry. To address this trade-off, we introduce two complementary losses:

- Context-Aware Loss ($\mathcal{L}^{ctx}$). As shown in Figure 4(i), we feed the full reference shape to the generator but calculate the loss only within the mask M . The localized supervision constrains the optimization to the masked region while leveraging the global context of the reference shape. This encourages the concept to maintain the semantic coherence between the masked region and the global structure.
- Context-Free Loss ($\mathcal{L}^{free}$). As shown in Figure 4(ii), we mask the user-specified region of the reference shape before feeding it into the generator. By requiring the model to reconstruct the masked region in the absence of surrounding context, we encourage the learned concept to be isolated from its neighbors and thus conceptually independent.

Below, we instantiate this region-wise concept learning framework for both geometry and appearance concepts.

Region-wise Geometry Concept Learning. For geometry, the reference data $\mathcal{D}$ refers to a latent feature grid X , obtained by encoding the voxelized reference shape. Given a binary voxel mask M , we downsample it to align with the latent feature grid resolution, obtaining M_{geo} . We initialize the text embedding C_{region} as S_{region} and the generator $\mathcal{G}$ as $\mathcal{G}_{geo}$. The specific losses are defined as:

$$\mathcal{L}_{geo}^{ctx} = \mathbb{E}_{t, X_t} [\|M_{geo} \odot (v_t - \mathcal{G}_{geo}(X_t, t, S_{region}))\|^2] \quad (5)$$

$$\text{and } \mathcal{L}_{geo}^{free} = \mathbb{E}_{t, X_t^M} [\|M_{geo} \odot (v_t - \mathcal{G}_{geo}(X_t^M, t, S_{region}))\|^2], \quad (6)$$

where X_t is the noisy grid derived from the latent X of the full reference shape, and X_t^M is derived from the latent X^M encoded from the masked reference shape.

$$\mathcal{L}_{geo}^{region} = \mathcal{L}_{geo}^{ctx} + \lambda_1 \mathcal{L}_{geo}^{free} \quad (7)$$

Region-wise Appearance Concept Learning. For appearance, we define the reference data as the SLAT representation $\mathcal{D} = (V, Z)$. Given a region specified by the user, we define a binary mask M_{app} on the sparse voxels V . We initialize the text embedding C_{region} as A_{region} and the generator $\mathcal{G}$ as $\mathcal{G}_{app}$ with the following objectives:

$$\mathcal{L}_{app}^{ctx} = \mathbb{E}_{t, Z_t, V} [\|M_{app} \odot (u_t - \mathcal{G}_{app}(Z_t, t, A_{region}, V))\|^2] \quad (8)$$



Fig. 5. Our method supports extracting concepts from global geometry, region-wise geometry, global appearance, and region-wise appearance, and composing the learned concepts with text to generate novel shapes.

and

$$\mathcal{L}_{app}^{free} = \mathbb{E}_{t, Z_t^M, V^M} [\|M_{app} \odot (u_t - \mathcal{G}_{app}(Z_t^M, t, A_{region}, V^M))\|^2], \quad (9)$$

where Z_t^M denotes the noisy latents derived from the masked appearance features $Z \odot M_{app}$, and $V^M = V \odot M_{app}$ represents the masked geometry containing only voxels within the masked region. The total region-wise appearance objective is defined as:

$$\mathcal{L}_{app}^{region} = \mathcal{L}_{app}^{ctx} + \lambda_2 \mathcal{L}_{app}^{free}. \quad (10)$$

The learned concept is represented by an optimized embedding and a fine-tuned generator: $(S'_{region}, \mathcal{G}'_{geo})$ for region-wise geometry concept and $(A'_{region}, \mathcal{G}'_{app})$ for region-wise appearance concept.

4.4 Shape Concept Inference

The learned concepts serve as reusable components during inference. By leveraging the decoupled designs of geometry and appearance concepts, our framework allows them to be applied independently, enabling flexible personalization with fine-grained control. Below, we describe the concept inference for geometry and appearance.

Geometry Concept Inference. We employ the learned geometry concept $(S', \mathcal{G}'_{geo})$ to synthesize new 3D shapes. Specifically, we construct a target text prompt y_{target} by combining the learned text embedding S' with new text prompts and feed the y_{target} into the fine-tuned generator $\mathcal{G}'_{geo}$ to produce a novel shape. By leveraging the geometric attributes encoded in $(\mathcal{G}'_{geo}, S')$, we can produce a shape follows the semantic controls of the text prompt while inheriting the geometric attributes of the reference shape.

Appearance Concept Inference. Given a generated geometry V , we employ the learned appearance concept $(A', \mathcal{G}'_{app})$ to synthesize its

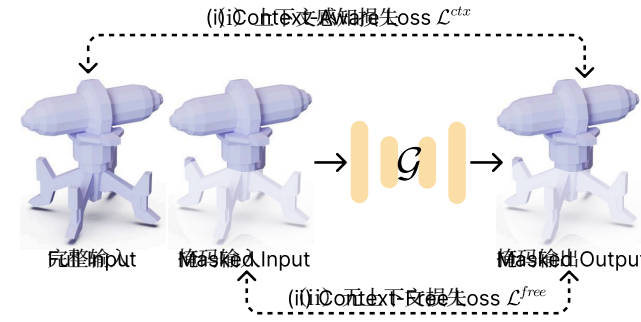


图 4. 区域级概念学习目标。 (i) 上下文感知损失使用完整参考图，并在掩码区域内计算损失。 (ii) 无上下文损失使用掩码输入，在掩码区域内计算损失。

而不是作为独立属性存在。相反，如果我们通过遮盖形状的其余部分来严格隔离该区域，概念可能会失去与全局几何之间的语义连贯性。为了解决这一权衡，我们引入了两个互补损失：

- 上下文感知损失 ($\mathcal{L}^{ctx}$)。如图 4(i) 所示，我们将完整参考形状输入生成器，但仅在掩码 M 内计算损失。局部监督将优化限制在掩码区域内，同时利用参考形状的全局上下文。这促使概念保持掩码区域与全局结构之间的语义连贯性。
- 无上下文损失 ($\mathcal{L}^{free}$)。如图 4(ii) 所示，我们先对参考形状中的用户指定区域进行掩码，然后将其输入生成器。通过要求模型在缺少周围上下文的情况下重建掩码区域，我们促使学习到的概念与其相邻部分相互隔离，从而在概念上保持独立。

下面，我们分别针对几何和外观概念实例化这一框架，开展区域级概念学习。

区域级几何概念学习。 对于几何，参考数据 $\mathcal{D}$ 指的是通过编码体素化参考形状获得的潜在特征网格 X 。给定二值体素掩码 M ，我们对其进行下采样，使其与潜在特征网格分辨率对齐，从而得到 M_{geo} 。我们将文本嵌入 C_{region} 初始化为 S_{region} ，并将生成器 $\mathcal{G}$ 初始化为 $\mathcal{G}_{geo}$ 。具体损失定义如下：

$$\mathcal{L}_{geo}^{ctx} = \mathbb{E}_{t, X_t} [\|M_{geo} \odot (v_t - \mathcal{G}_{geo}(X_t, t, S_{region}))\|^2] \quad (5)$$

$$\text{and } \mathcal{L}_{geo}^{free} = \mathbb{E}_{t, X_t^M} [\|M_{geo} \odot (v_t - \mathcal{G}_{geo}(X_t^M, t, S_{region}))\|^2], \quad (6)$$

其中， X_t 是由完整参考形状的潜变量 X 得到的噪声网格，而 X_t^M 则由掩码参考形状编码得到的潜变量 X^M 生成。

$$\mathcal{L}_{geo}^{region} = \mathcal{L}_{geo}^{ctx} + \lambda_1 \mathcal{L}_{geo}^{free} \quad (7)$$

区域级外观概念学习。 对于外观，我们将参考数据定义为 SLAT 表示 $\mathcal{D} = (V, Z)$ 。给定用户指定的区域，我们在稀疏体素 V 上定义二值掩码 M_{app} 。我们将文本嵌入 C_{region} 初始化为 A_{region} ，并将生成器 $\mathcal{G}$ 初始化为 $\mathcal{G}_{app}$ ，其目标如下：

$$\mathcal{L}_{app}^{ctx} = \mathbb{E}_{t, Z_t, V} [\|M_{app} \odot (u_t - \mathcal{G}_{app}(Z_t, t, A_{region}, V))\|^2] \quad (8)$$



图5. 我们的方法支持从全局几何、区域几何结构、全局外观和区域外观中提取概念，并将学习到的概念与文本结合，以生成新颖形状。

and

$$\mathcal{L}_{app}^{free} = \mathbb{E}_{t, Z_t^M, V^M} [\|M_{app} \odot (u_t - \mathcal{G}_{app}(Z_t^M, t, A_{region}, V^M))\|^2], \quad (9)$$

其中 Z_t^M 表示由掩码外观特征 $Z \odot M_{app}$ 得到的噪声潜变量， $V^M = V \odot M_{app}$ 表示仅包含掩码区域内体素的掩码几何。总区域级外观目标定义为：

$$\mathcal{L}_{app}^{region} = \mathcal{L}_{app}^{ctx} + \lambda_2 \mathcal{L}_{app}^{free}. \quad (10)$$

学习到的概念由优化嵌入和微调生成器表示： $(S'_{region}, \mathcal{G}'_{geo})$ 用于区域级几何概念， $(A'_{region}, \mathcal{G}'_{app})$ 用于区域级外观概念。

4.4 形状概念推理

学习到的概念在推理过程中充当可复用组件。借助几何概念与外观概念的解耦设计，我们的框架允许独立应用二者，从而实现具有细粒度控制的灵活个性化。下面，我们将介绍几何和外观的概念推理。

几何概念推理。 我们使用学习到的几何概念 $(S', \mathcal{G}'_{geo})$ 来合成新的三维形状。具体而言，我们通过将学习到的文本嵌入 S' 与新的文本提示结合，构建目标文本提示 y_{target} ，并将其 y_{target} 输入微调生成器 $\mathcal{G}'_{geo}$ ，以生成新颖形状。利用 $(\mathcal{G}'_{geo}, S')$ 中编码的几何属性，我们可以生成一种遵循文本提示的语义控制、同时继承参考形状几何属性的形状。

外观概念推理。 给定生成的几何体 V ，我们运用所学习的外观概念 $(A', \mathcal{G}'_{app})$ 来合成其

Table 1. Quantitative comparisons for geometry concept. Our method surpasses existing baselines across all quantitative metrics and user study results, demonstrating the effectiveness of our approach.

Method	KID ↓	CLIP-R ↑	QS ↑	AS ↑	TS ↑
Qwen-Edit [Wu et al. 2025a]	0.0088	0.778	3.87	3.13	3.62
Edit360 [Huang et al. 2025]	0.0082	0.644	3.43	2.44	3.53
Coin3D [Dong et al. 2024]	0.0176	0.544	2.45	1.81	2.39
VoxHammer [Li et al. 2026]	0.0094	0.200	3.44	4.11	1.23
StyleSculptor [Qu et al. 2025]	0.0197	0.589	3.88	1.30	4.27
Ours	0.0064	0.833	4.41	4.15	4.31

Table 2. Quantitative comparisons for appearance concept. Our method outperforms all baselines in both attribute preservation and overall visual quality, as demonstrated by both metric evaluations and user studies.

Method	ArtFID ↓	FID ↓	QS ↑	MS ↑
Paint3D [Zeng et al. 2024]	29.16	280.5	3.42	2.91
StyleTex [Xie et al. 2024]	30.42	266.5	2.76	2.25
StyleSculptor [Qu et al. 2025]	27.50	258.5	3.25	2.92
Ours	23.45	235.9	4.67	4.37

appearance. The fine-tuned appearance generator $\mathcal{G}'_{app}$ produces the appearance latent Z , conditioned on the geometry V and the appearance text embedding A' . Finally, the structured latents (V, Z) are decoded into a final 3D asset. By utilizing the appearance attributes encoded within $(A', \mathcal{G}'_{app})$, the produced 3D asset maintains an appearance that is consistent with the original reference.

Geometry/Appearance Concepts Composition. As Figures 1 and 5 show, we support composing learned geometry and appearance concepts, learned from either global or region-wise mechanism, to synthesize new shapes. We first apply the learned geometry concept composed with the text prompt to generate the shape geometry. Then, we apply the appearance concept with the generated geometry to synthesize its appearance. This design enables flexible concepts composition and provides users with fine-grained controllability.

5 Results and Experiments

5.1 Experiment Settings

To evaluate the effectiveness of our method, we compare it with various baselines for both geometry and appearance concepts.

Geometry Concept. To evaluate our method under this new task, we conduct a comprehensive evaluation by adapting five representative methods from closely related fields: (i) Qwen-Edit [Wu et al. 2025a], (ii) Edit360 [Huang et al. 2025], (iii) StyleSculptor [Qu et al. 2025], (iv) Coin3D [Dong et al. 2024], and (v) VoxHammer [Li et al. 2026]. Qwen-Edit and Edit360 perform image personalization followed by 3D reconstruction [Wang et al. 2021; Xiang et al. 2025]. VoxHammer targets text-guided shape editing, whereas Coin3D combines coarse geometric proxies with text for generation. StyleSculptor requires a target shape for 3D stylization; we generate it from text using TRELIS. For a comprehensive evaluation, we introduce a benchmark of 30 reference shapes from Objaverse-XL [Deitke et al. 2023a]. Each reference is paired with three distinct text prompts targeting different categories, establishing a total of 90 evaluation cases. More details about the geometry concept benchmark are provided in supplementary Section A.1.

Appearance Concept. To evaluate our method for the new appearance concept task, we perform a comprehensive evaluation against three methods adapted from closely related fields: (i) Paint3D [Zeng et al. 2024], (ii) StyleTex [Xie et al. 2024], and (iii) StyleSculptor [Qu et al. 2025]. Since Paint3D and StyleTex apply appearance from an image to a shape, we adapt them by rendering the reference appearance shape into an input image. For StyleSculptor, it requires source and target images, so we render both the reference appearance shape and the geometry shape. For a fair evaluation, we constructed a benchmark of 30 pairs from Objaverse-XL [Deitke et al. 2023a], where each pair consists of a reference appearance shape and a geometry shape. More details about the appearance concept benchmark are provided in supplementary Section A.2.

5.2 Quantitative Comparison

Evaluation Metrics. To evaluate geometry concept, we first assess the visual quality of the generated shapes using Kernel Inception Distance (KID). Following [Liu et al. 2024; Raj et al. 2023], we utilize CLIP R-Precision (CLIP-R) to evaluate the text alignment. For appearance concept, we adopt ArtFID [Wright and Ommer 2022] to assess appearance fidelity between the reference and generated results. Then, we follow [Qu et al. 2025] to use Fréchet Inception Distance (FID) to measure the visual quality of the generated appearance.

User Study. Following [Hu* et al. 2023; Hu et al. 2024b], we conducted a user study with 10 participants to evaluate geometry and appearance concepts. Participants rated results on a Likert scale from 0 (worst) to 5 (best). For geometry, evaluations focused on: (i) Quality Score (QS), reflecting visual quality; (ii) Attribute Score (AS), assessing the preservation of geometric attributes; and (iii) Text Score (TS), evaluating how well the generated shapes follow the text. For appearance, participants evaluated: (i) Quality Score (QS); and (ii) Matching Score (MS), measuring the similarity between the generated appearance and the reference.

Table 1 presents the quantitative results for geometry concept. Across all metrics, our method outperforms prior approaches: it produces higher-quality geometry (higher QS, lower KID), better preserves reference attributes (higher CLIP-R, AS), and follows text control more faithfully (higher CLIP-R, TS). Table 2 reports the results for appearance concept. Our method achieves the best performance, with stronger appearance preservation (lower ArtFID, higher MS) and higher-quality generations (lower FID, higher QS).

5.3 Qualitative Comparison

Geometry Concept. Figure 8 presents visual comparisons for the geometry concept. As shown in second row, our method can produce results that preserve geometric attributes, *e.g.*, vertical slats, while adhere to the text control. In contrast, baselines either discard geometric attributes or fail to match the text.

Appearance Concept. Figure 9 provides visual comparisons for appearance concept. Our method better preserves the appearance attribute of the reference than baselines. As shown in row 1 (left), our method retains the bridge’s blue and black pattern, whereas others alter the color distribution.

More comparisons are showed in supplementary Sections B & C.

表 1. 几何概念的定量比较。我们的方法在所有定量指标和用户研究结果上均优于现有基线方法，证明了我们方法的有效性。

方法	KID ↓	CLIP-R ↑	QS ↑	AS ↑	TS ↑
Qwen-Edit [Wu 等, 2025a]	0.0088	0.778	3.87	3.13	3.62
编辑360 [Huang 等 2025]	0.0082	0.644	3.43	2.44	3.53
Coin3D [Dong 等人 2024]	0.0176	0.544	2.45	1.81	2.39
VoxHammer [Li 等人 2026]	0.0094	0.200	3.44	4.11	1.23
StyleSculptor [Qu 等人 2025]	0.0197	0.589	3.88	1.30	4.27
Ours	0.0064	0.833	4.41	4.15	4.31

表 2. 外观概念的定量比较。正如指标评估和用户研究所展示的，我们的方法在属性保持和整体视觉质量方面均优于所有基线方法。

方法	ArtFID ↓	FID ↓	QS ↑	MS ↑
Paint3D [Zeng 等人 2024]	29.16	280.5	3.42	2.91
StyleTex [Xie 等人 2024]	30.42	266.5	2.76	2.25
StyleSculptor [Qu 等人 2025]	27.50	258.5	3.25	2.92
Ours	23.45	235.9	4.67	4.37

外观。经过微调的外观生成器 $\mathcal{G}'_{app}$ 生成外观潜变量 Z ，其条件为几何体 V 和外观文本嵌入 A' 。最后，将结构化潜变量 (V, Z) 解码为最终的3D资产。通过利用编码在 $(A', \mathcal{G}'_{app})$ 中的外观属性，生成的3D资产保持了与原始参考图一致的外观。

几何/外观概念组合。如图 1 和图 5 所示，我们支持组合从全局机制或区域级机制中学习到的几何概念和外观概念，以合成新的形状。我们首先将学习到的几何概念与文本提示结合，用于生成形状几何。然后，我们将外观概念与生成的几何体结合，以合成其外观。这种设计实现了灵活的概念组合，并为用户提供了细粒度的可控性。

5 结果与实验

5.1 实验设置

为评估我们方法的有效性，我们针对几何概念与外观概念，分别将其与多种基线方法进行比较。

几何概念。为在这一新任务下评估我们的方法，我们通过改编来自密切相关领域的五种代表性方法开展了全面评估：(i)) Qwen-Edit [Wu 等, 2025a], (ii) Edit360 [Huang 等, 2025], (iii) StyleSculptor [Qu 等, 2025], (iv) Coin3D [Dong 等, 2024], 以及 (v) VoxHammer [Li 等, 2026]。Qwen-Edit 和 Edit360 先进行图像个性化，再进行三维重建 [Wang 等, 2021; Xiang 等, 2025]。VoxHammer 面向文本引导的形状编辑，而 Coin3D 将粗略几何代理与文本相结合进行生成。StyleSculptor 需要一个目标形状来进行三维风格化；我们使用 TRELIS 根据文本生成该形状。为进行全面评估，我们引入了一个包含来自 Objaverse-XL [Deitke 等, 2023a]的 30 个参考形状的基准测试集。每个参考形状都与三个针对不同类别的文本提示配对，共构成 90 个评估案例。有关几何概念基准测试集的更多细节，请参见补充材料第 A.1 节。

外观概念。为评估我们的方法在新外观概念任务上的表现，我们针对三种改编自密切相关领域的方法开展了全面评估：(i) Paint3D [Zeng 等, 2024], (ii) StyleTex [Xie 等, 2024], 以及 (iii) StyleSculptor [Qu 等, 2025]。由于 Paint3D 和 StyleTex 是将图像中的外观应用到形状上，我们通过将参考外观形状渲染为输入图像来对其进行改编。对于 StyleSculptor，它需要源图像和目标图像，因此我们分别渲染参考外观形状和几何形状。为确保评估公平，我们构建了一个包含来自 Objaverse-XL [Deitke 等, 2023a]的 30 对样本的基准测试集，其中每对样本由一个参考外观形状和一个几何形状组成。有关外观概念基准测试集的更多细节，请参见补充材料第 A.2 节。

5.2 定量比较

评估指标。为评估几何概念，我们首先使用核Inception距离（KID）评估生成形状的视觉质量。根据 [Liu 等人 2024; Raj 等人 2023]，我们采用 CLIPR-Precision (CLIP-R) 评估文本对齐度。对于外观概念，我们采用 ArtFID [Wright 和 Ommer 2022] 评估参考图与生成结果之间的外观保真度。随后，我们遵循 [Qu 等人 2025] 的做法，使用 Fréchet Inception距离（FID）衡量生成外观的视觉质量。

用户研究。根据 [Hu* 等人 2023; Hu 等人 2024b]，我们邀请 10 名参与者开展用户研究，以评估几何概念与外观概念。参与者使用 0（最差）至 5（最佳）的李克特量表对结果进行评分。对于几何概念，评估重点包括：(i) 质量分数（QS），反映视觉质量；(ii) 属性分数（AS），评估几何属性的保留情况；以及 (iii) 文本分数（TS），评估生成形状遵循文本描述的程度。对于外观概念，参与者评估：(i) 质量分数（QS）；以及 (ii) 匹配分数（MS），衡量生成外观与参考图之间的相似度。

表 1 展示了几何概念的定量结果。在所有指标上，我们的方法都优于先前方法：它能够生成更高质量的几何结构（更高的 QS、更低的 KID），更好地保留参考属性（更高的 CLIP-R、AS），并且更忠实地遵循文本控制（更高的 CLIP-R、TS）。表 2 报告了外观概念的结果。我们的方法取得了最佳性能，具有更强的外观保留能力（更低的 ArtFID、更高的 MS）和更高质量的生成结果（更低的 FID、更高的 QS）。

5.3 定性比较

几何概念。图 8 展示了几何概念的视觉比较。如第二行所示，我们的方法能够生成保留几何属性的结果，例如垂直条板，同时遵循文本控制。相比之下，基线方法要么舍弃几何属性，要么无法匹配文本。

外观概念。图 9 展示了外观概念的视觉比较。与基线方法相比，我们的方法能够更好地保留参考图的外观属性。如第 1 行（左侧）所示，我们的方法保留了桥梁的蓝黑色图案，而其他方法则改变了颜色分布。

更多比较结果请参见补充材料 B 和 C 节。

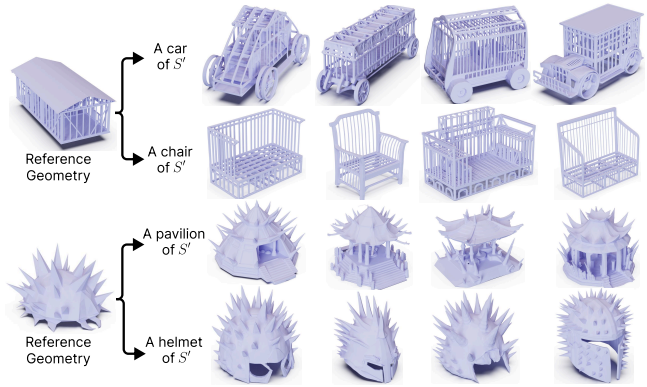


Fig. 6. Visual results for global geometry concept variations. Our method generates diverse shapes that preserve the same geometric attributes, *e.g.*, vertical slats or spikes, while consistently following the same text controls.

5.4 More Visual Results

We provide additional visual results for our method. We first present results for the global geometry concept: Figure 6 illustrates the visual result for concept-level shape variations with the same text, while Figure 10 demonstrates the synthesis of novel shapes using different text prompts. Next, Figure 11 displays results for the global appearance concept. We then show results for region-wise concept: Figure 12 provides additional visualizations for region-wise geometry concept, and Figure 13 presents visual results of the region-wise appearance concept. More visual results are provided in supplementary material Sections D, E, F, and G.

5.5 Ablation Study

We first perform ablation to validate progressive optimization strategy. When relying solely on embedding optimization, the method struggles to capture fine-grained geometric and appearance attributes. As Figure 7(c) shows, it fails to preserve the geometric attributes and generates a generic robot. For appearance, it cannot reproduce the black-and-white pattern, resulting in a generic giraffe-like texture. Yet, using only model fine-tuning causes the model to overfit and lose text control. As Figure 7(d) demonstrates, the geometry fails to follow the text control, while the appearance mimics white colors rather than the black-white patterns. In contrast, our full pipeline effectively balances these trade-offs, ensuring both attribute preservation and text control; see Figure 7(b).

We next ablate the two complementary losses in our region-wise objective. First, we remove the context-aware loss $\mathcal{L}^{ctx}$ for ablation. As shown in Figure 7 (g), removing $\mathcal{L}^{ctx}$ degrades visual coherency. While the geometry concept captures vertical slats, it appears fragmented, and the appearance concept fails to retain the pink-and-white pattern. Second, removing $\mathcal{L}^{free}$ makes the model fail to preserve attributes. As Figure 7(h) shows, geometry is generated solely by the text while generated appearance struggles to maintain the pink-and-white pattern.

Table 3 shows quantitative ablation studies. Removing either embedding optimization or model fine-tuning hinders both geometry and appearance concept learning.

Table 3. Quantitative comparison of our full pipeline against ablated cases.

Method	Geometry		Appearance	
	KID ↓	CLIP-R ↑	ArtFID ↓	FID ↓
Ours w/o Embedding Opt.	0.0075	0.567	29.21	276.03
Ours w/o Model Fine-tuning	0.0117	0.489	31.82	302.76
Full pipeline	0.0064	0.842	23.45	235.9

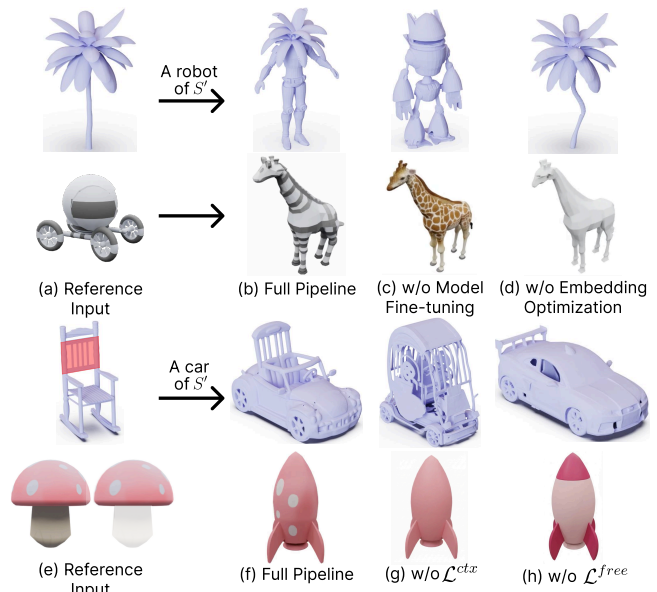


Fig. 7. Visual results for the ablation studies. (c) Removing model fine-tuning weakens attribute capture, while (d) removing embedding optimization leads to overfitting. For region-wise learning, (g) removing context-aware loss reduces visual coherence, whereas (h) removing the context-free loss hinders the model from capturing the reference attributes.

6 Conclusions

We have presented a notion of 3D personalization that shifts the focus from instance-level control to concept-level abstraction. Rather than cloning, deforming, or editing a reference shape, our approach treats personalization as the extraction of reusable geometric and appearance concepts that can be recomposed across categories. This formulation departs from target-driven pipelines that operate by modifying an existing shape, and instead enables reference-guided generation without prescribing any target instance. Conceptually, this places 3D personalization closer to language-level creativity, where abstract attributes are learned from examples and flexibly combined with semantic intent, rather than framing it as yet another geometry editing or localized shape manipulation technique. By operating at transferable concept level, the method supports a more compositional and open-ended paradigm for 3D content creation.

Building on this notion, we showed that framing personalization around concept extraction leads to a more compositional view of 3D generation. Concepts could be learned at different scales, extracted locally or globally, and combined with one another and with text in a flexible manner. Rather than producing a single edited result, this opens the door to more modular and expressive ways of creating

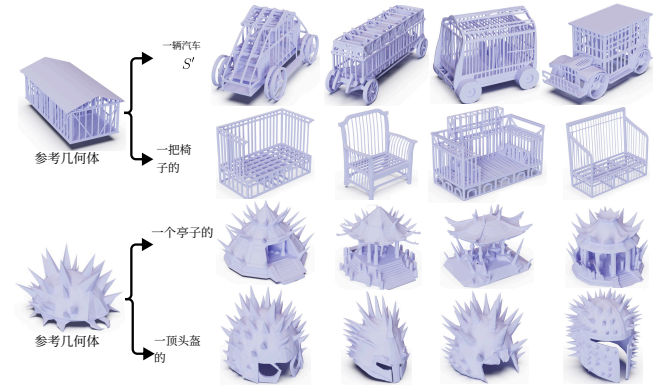


图6. 全局几何概念变化的视觉结果。我们的方法生成了多样形状，同时保持相同的几何属性，例如垂直条板或尖刺，并始终遵循相同的文本控制。

5.4 更多视觉结果

我们为方法提供了额外的视觉结果。首先，我们展示全局几何概念的结果：图 6 展示了使用相同文本时概念级形状变化的视觉结果，而图 10 展示了使用不同文本提示合成新颖形状的结果。接下来，图 11 展示了全局外观概念的结果。随后，我们展示区域级概念的结果：图 12 提供了区域级几何概念的更多可视化结果，图 13 展示了区域级外观概念的视觉结果。更多视觉结果请参见补充材料 D、E、F 和 G 节。

5.5 消融实验

我们首先进行消融实验，以验证渐进式优化策略。仅依赖嵌入优化时，该方法难以捕捉细粒度的几何与外观属性。如图7(c)所示，它无法保持几何属性，并生成了一个通用机器人。在外观方面，它无法复现黑白图案，resulting in a generic giraffe-like texture。另一方面，仅使用模型微调会导致模型过拟合并失去文本控制。如图7(d)所示，几何形状无法遵循文本控制，而外观模仿的是白色，而非黑白图案。相比之下，我们的完整流程有效平衡了这些权衡，确保属性保持与文本控制兼备；见图7(b)。

接下来，我们对区域级目标中的两个互补损失进行消融。首先，我们移除上下文感知损失 $\mathcal{L}^{ctx}$ 以进行消融。如图7(g)所示，移除 $\mathcal{L}^{ctx}$ 会降低视觉连贯性。尽管几何概念捕捉到了垂直条板，但其呈现得较为碎片化，且外观概念无法保留粉白图案。其次，移除 $\mathcal{L}^{free}$ 会导致模型无法保留属性。如图7(h)所示，几何完全由文本生成，而生成的外观难以保持粉白图案。

表3展示了定量消融研究。移除嵌入优化或模型微调中的任一项，都会阻碍几何以及外观概念学习。

表3. 我们的完整流程与消融案例的定量比较。

方法	几何		外观	
	KID ↓	CLIP-R ↑	ArtFID ↓	FID ↓
我们的方法（无嵌入优化）	0.0075	0.567	29.21	276.03
我们的方法（无模型微调）	0.0117	0.489	31.82	302.76
完整流程	0.0064	0.842	23.45	235.9

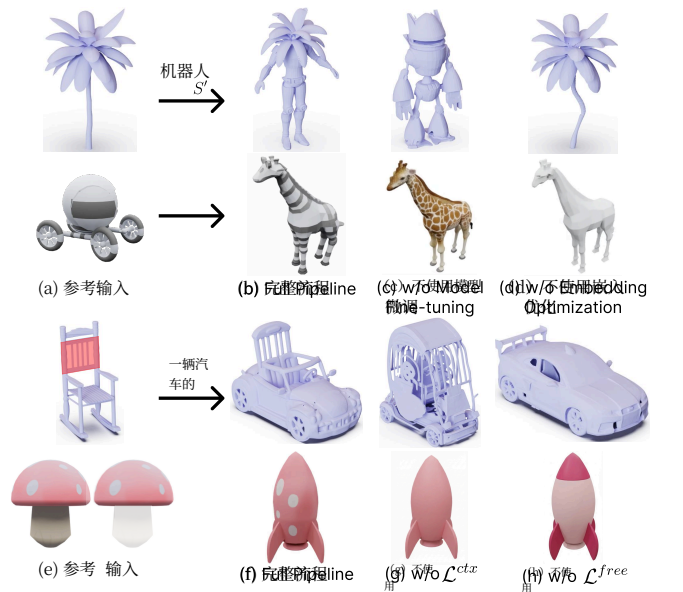


图7. 消融研究的视觉结果。（c）移除模型微调会削弱属性捕捉，而（d）移除嵌入优化会导致过拟合。对于区域级学习，（g）移除上下文感知损失会降低视觉连贯性，而（h）移除无上下文损失会妨碍模型捕捉参考属性。

6 结论

我们提出了三维个性化这一概念，将重点从实例级控制转向概念级抽象。我们的方法并非克隆、变形或编辑参考形状，而是将个性化视为提取可复用的几何概念和外观概念，使其能够跨类别重新组合。这一表述不同于通过修改现有形状来实现的目标驱动管线，而是支持参考引导生成，无需预先指定任何目标实例。从概念上看，这使三维个性化更接近语言层面的创造力：从示例中学习抽象属性，并根据语义意图灵活组合，而不是将其视为又一种几何编辑或局部形状操控技术。通过在可迁移的概念层面进行操作，该方法为三维内容创作提供了一种更具组合性和开放性的范式。

基于这一理念，我们展示了：围绕概念提取来构建个性化，能够带来对三维生成更具组合性的理解。概念可以在不同尺度上习得，可以进行局部或全局提取，还能以灵活的方式彼此组合，并与文本结合。与其生成单一的编辑结果，这种方式为更加模块化且富有表现力的创作方式打开了大门

3D content, where new designs can emerge from the reuse and recombination of learned ideas.

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三维内容，其中新的设计可以通过对习得想法的再利用与重组而产生。

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Fig. 8. Visual comparisons of geometry concept with Qwen-Edit [Wu et al. 2025a], Edit360 [Huang et al. 2025], StyleSculptor [Qu et al. 2025], Coin3D [Dong et al. 2024], and VoxHammer [Li et al. 2026]. Our results better preserve geometric attributes of the reference while more faithfully following text controls.



Fig. 9. Visual comparisons of appearance concept with Paint3D [Zeng et al. 2024], StyleTex [Xie et al. 2024], StyleSculptor [Qu et al. 2025]. Our method preserves the appearance attributes of the reference, while other approaches struggle with it.

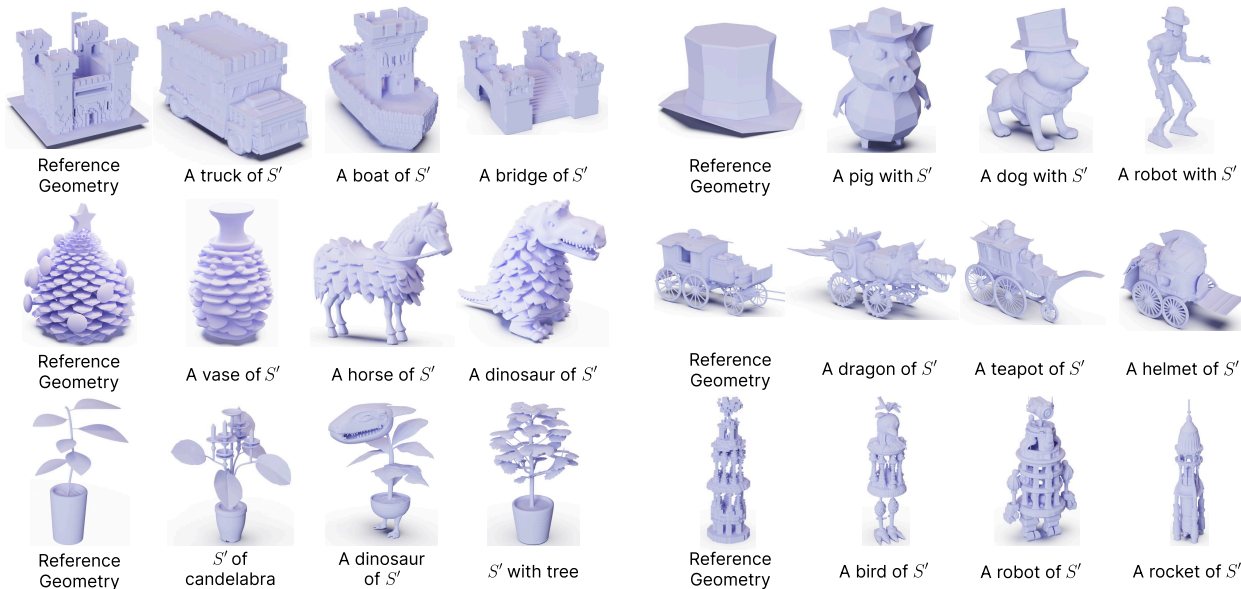


Fig. 10. Visual results for global geometry concept. Our method learns a global geometry concept from a reference shape and composes it with different text prompts to synthesize diverse shapes that consistently preserve the reference geometric attributes.



图 8. 使用 Qwen-Edit [Wu 等, 2025a], Edit360 [Huang 等 2025], StyleSculptor [Qu 等 2025], Coin3D [Dong 等 2024],以及 VoxHammer [Li 等 2026] 进行几何概念的视觉对比。我们的方法能够更好地保留参考图的几何属性，同时更忠实地遵循文本控制。



图 9. 使用 Paint3D [Zeng 等 2024]、StyleTex [Xie 等 2024], 和 StyleSculptor [Qu 等 2025]进行外观概念的视觉对比。我们的方法能够保留参考图的外观属性，而其他方法则难以做到这一点。

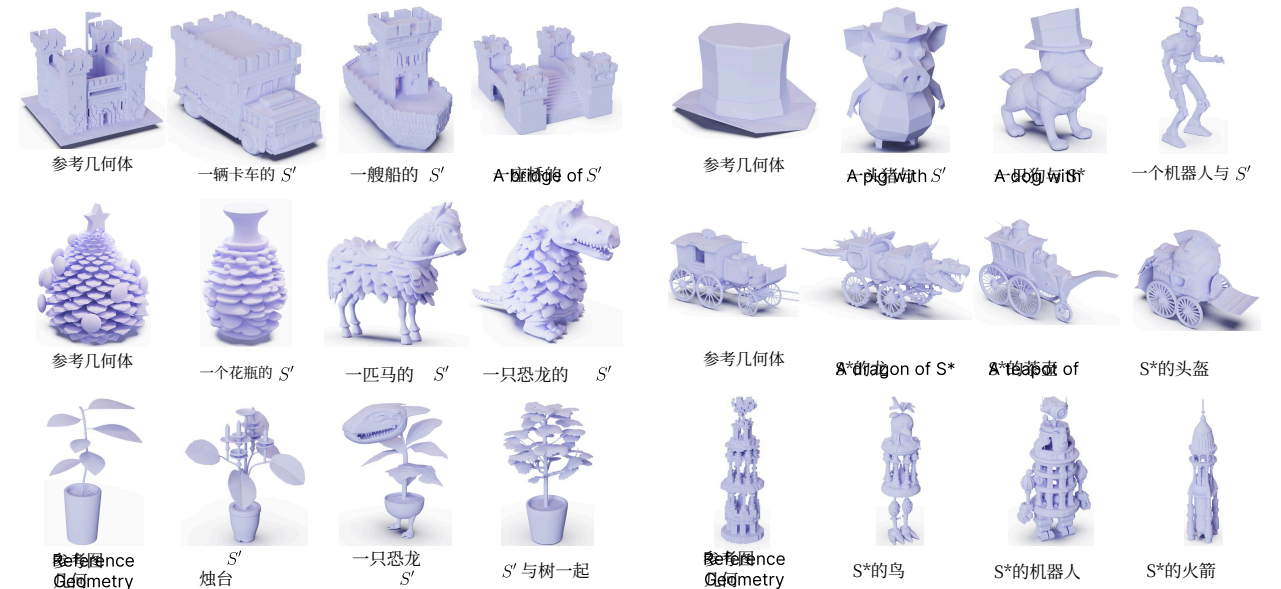


图 10. 全局几何概念的视觉结果。我们的方法从参考形状中学习全局几何概念，并将其与不同的文本提示结合，以合成多样形状，同时始终保留参考形状的几何属性。

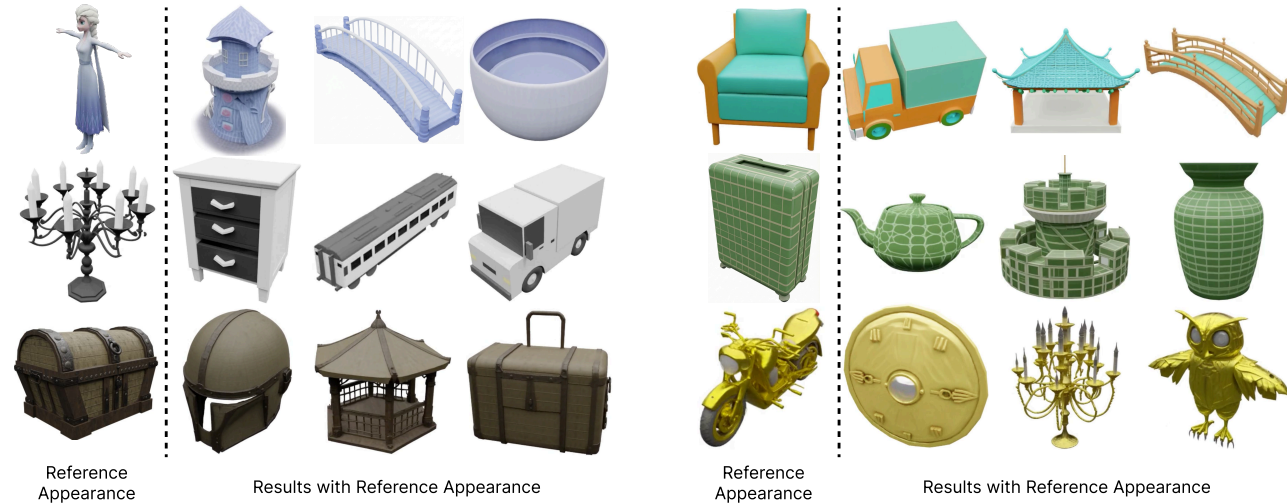


Fig. 11. Visual results for global appearance concept. By extracting global appearance concept, our approach produces multiple high-fidelity results that consistently reflect the reference appearance attributes.

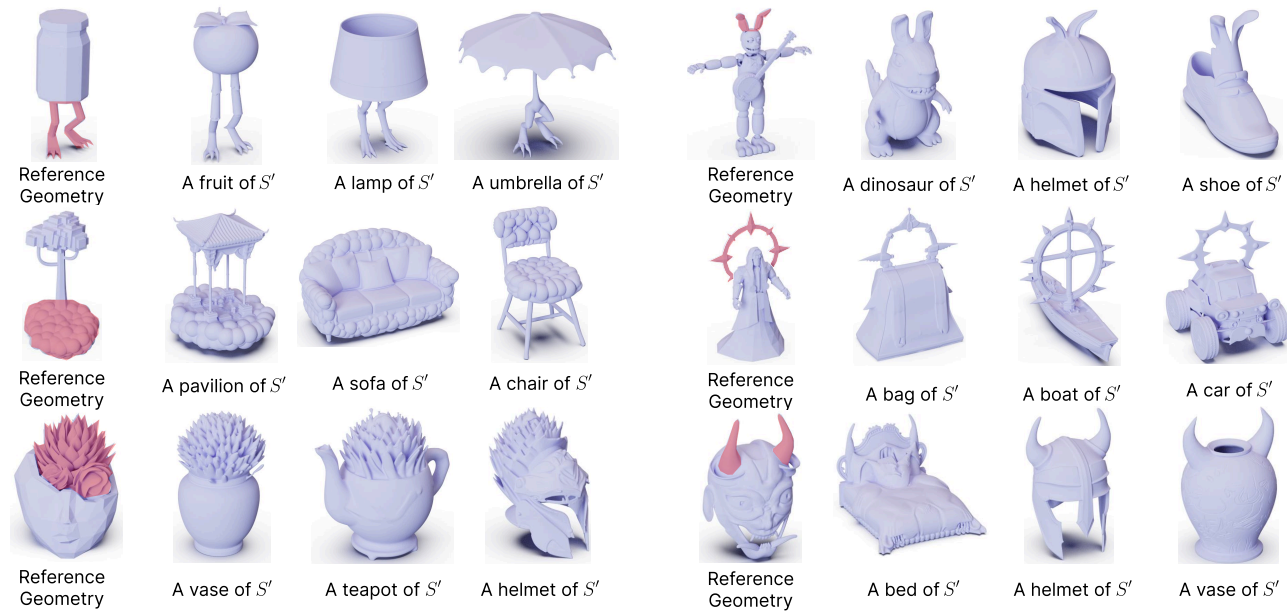


Fig. 12. Visual results for region-wise geometry concept. Our method enables extract the geometry concept from a user-specified region (marked in red), and compose it with text to generate novel shapes.

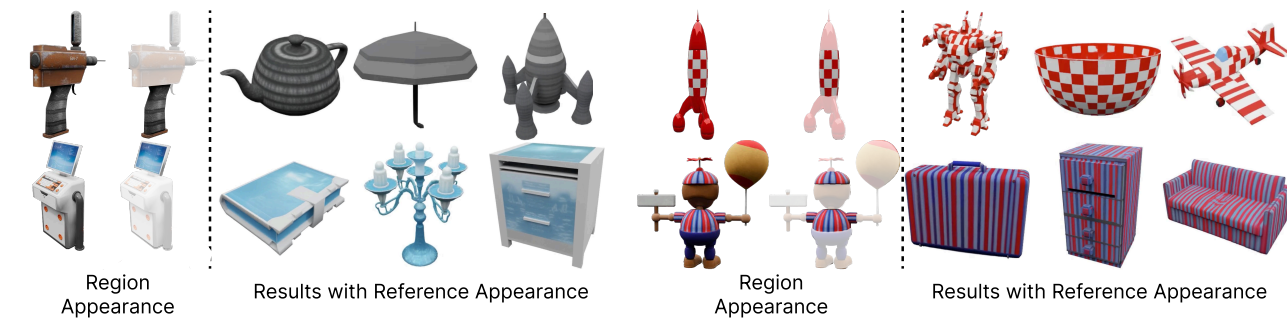


Fig. 13. Visual results for region-wise appearance concept. Our method enables extracting appearance concept from a localized region and reuse it to generate results that consistently inherit the reference appearance.

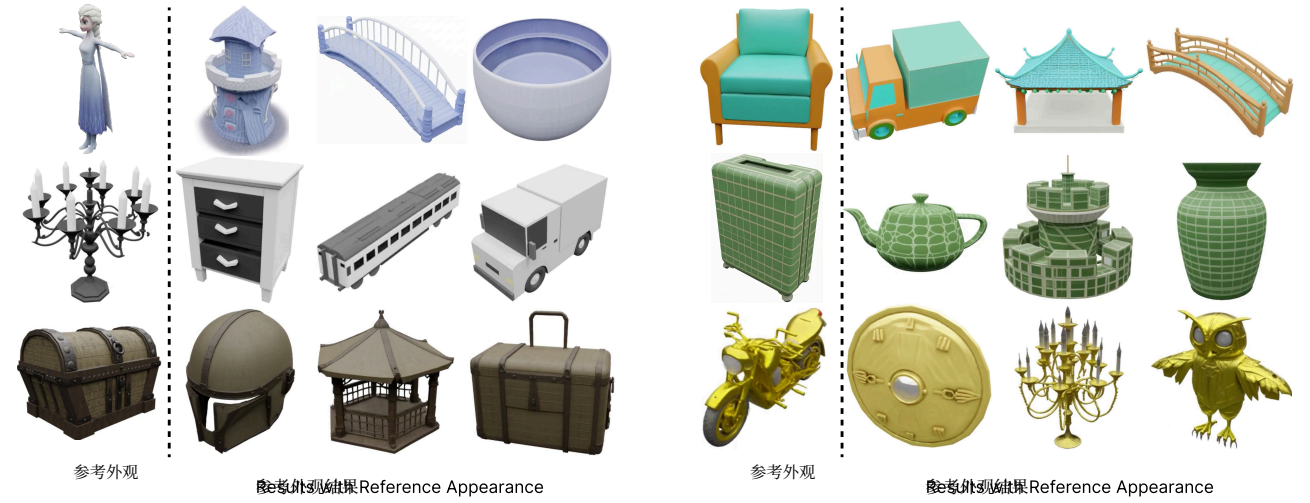


图11. 全局外观概念的视觉结果。通过提取全局外观概念，我们的方法能够生成多个高保真结果，并始终体现参考外观属性。



图12. 区域级几何概念的视觉结果。我们的方法能够从用户指定区域（以红色标记）中提取几何概念，并将其与文本结合以生成新颖形状。

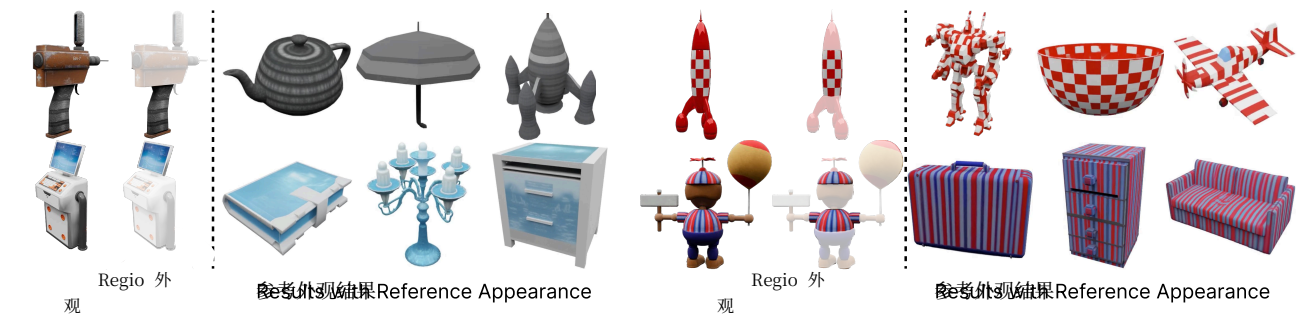


图13. 区域级外观概念的视觉结果。我们的方法能够从局部区域提取外观概念，并再利用该概念生成始终继承参考外观的结果。